Cristobal, Hailey Jade **BSIT 2-1** **OOP**

Midterm Activity #2 - Operators

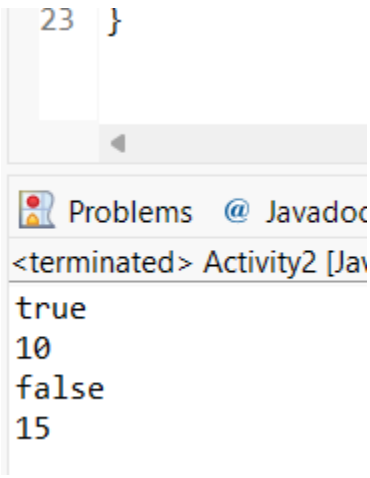
I. Part One

1. Create a Java program using bitwise operators that will produce the output: true, 10, false, 15.

SC:

```
1  package one;
2
3  public class Activity2{
4
5      public static void main(String[] args){
6
7          int x = 12;  // 1100
8          int y = 10;  // 1010
9
10         boolean result1 = ((x & y) > 0);   // 1000 = 8 > 0
11         System.out.println(result1);
12
13         int result2 = y & x;   // 8
14         result2 = y | 0;       // 10 (still computed)
15         System.out.println(result2);
16
17         boolean result3 = ((x ^ x) > 0);  // 0
18         System.out.println(result3);
19
20         int result4 = 7 | 8;
21         System.out.println(result4);
22     }
23 }
```

OUTPUT:



2. Create a Java program using the ternary operator that will produce the output: true, false, 25, 30.

SC:

```java
package one;

public class Activity2{

    public static void main(String[] args){

        int a = 20;
        int b = 15;

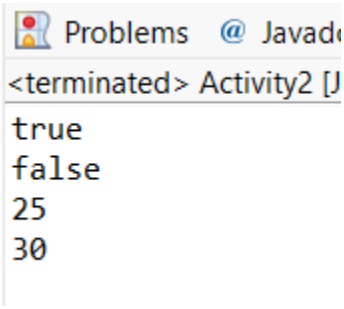
        boolean result1 = (a > b) ? true : false;
        System.out.println(result1);

        boolean result2 = (a < b) ? true : false;
        System.out.println(result2);

        int result3 = (a > b) ? 25 : 10;
        System.out.println(result3);

        int result4 = (b < a) ? 30 : 5;
        System.out.println(result4);
    }
}
```

OUTPUT:



3. Create a Java program using assignment operators that will produce the output: Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.

SC:

```java
package one;

public class Activity2{

    public static void main(String[] args){

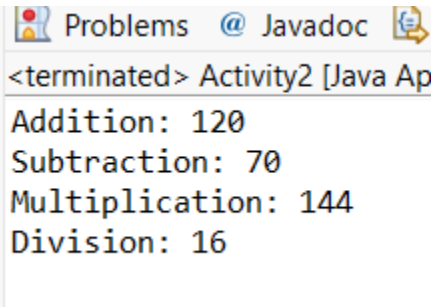
        int num1 = 100;
        num1 += 20;
        System.out.println("Addition: " + num1);

        int num2 = 100;
        num2 -= 30;
        System.out.println("Subtraction: " + num2);

        int num3 = 12;
        num3 *= 12;
        System.out.println("Multiplication: " + num3);

        int num4 = 64;
        num4 /= 4;
        System.out.println("Division: " + num4);
    }
}
```

OUTPUT:



II. Part Two

SC:

```java
package one;

public class Activity2{

    public static void main(String[] args){

        // increment case
        int p = 3;
        int tricky = p++ + ++p;   // 3 + 5 = 8
        System.out.println(tricky);

        // arithmetic + assignment
        int q = 14;
        q *= 3;   // 42
        System.out.println(q);

        // bitwise shift
        int r = 2;
        int shiftResult = r << 3;   // 16
        System.out.println(shiftResult);

        // modulus
        int s = 29 % 6;   // 5
        System.out.println(s);

        // relational + logical
        boolean check1 = (tricky > q) && (shiftResult > s);
        System.out.println(check1);

        // logical with NOT
        boolean check2 = !(q == shiftResult);
        System.out.println(check2);
    }
}
```

OUTPUT:

